

Neck-loop, audio-streamer for hearing-aid users

ClearSounds' Quattro XS is an amplified, Bluetooth neck-loop with advanced audio distribution profile (A2DP). Perfect for hearing-aid users, it ensures that they hear perfectly in any environment. Transmission of sound from a Bluetooth-enabled mobile phone, directly through the T-coil of a hearing aid, reduces or eliminates interference between cellular phones and hearing aids or cochlear processors. It also tries to shut off background noise to a large extent. The device provides voice-controls for volume, amplification, dialling, etc.

Inside: Quattro XS features a binaural design. It includes an amplifier and removable Bluetooth microphone, as well as a user-friendly noise-cancelling neck-loop. A push-button slider is used to cinch the neck-loop for closer microphone placement. The device uses a universal micro-USB for fast charging and includes an attachable Bluetooth mini-microphone transmitter, intercom features, talking caller ID, and sophisticated profile signal handling between the Bluetooth device and neck-loop signal interface. The headset jack accommodates audio headsets for non-T-coil use. Quattro offers 30dB adjustable amplification, and supports various profiles including hands-free, headset, A2DP and audio/video remote control profile. It is compatible with standard Bluetooth Version 2.1 + EDR devices, and has a range of 9 metres (30 feet). It uses a Lithium-Ion battery that offers a talk time of up to three hours and standby time of five days. It can be charged through USB cable, car adaptor or AC adaptor, all of which are included in the package.



Control battery-powered devices wirelessly

Power management solutions are all the rage. We keep hearing about controllers for air-conditioners, lighting systems, computers and so on. But what about smaller, battery-powered devices? Is it possible to turn off your radio through your mobile phone? Yes, Tethercell, an innovative product developed by Trey Madhyastha and Kellan O'Connor, veterans of the aerospace industry, lets you control the gadget you want. Just remove one of the AA-size batteries from the gadget and replace it with a Tethercell adaptor, which is a little app-enabled device with the dimensions of an AA-size battery. You can now remotely control the gadget through an app running on your iOS or Android device. Tethercell will be available commercially from June 2013.

Inside: Tethercell contains a lot of cutting-edge electronics based upon the TI CC2540 microcontroller. It also contains a current-sensing op-amp comparator, temperature sensor, n-channel MOSFET (capable of switching up to 5A), 1.5-3V boost converter and embedded 8051 microcontroller. The device is powered by a single AAA-size battery. Powering the on-board Bluetooth radio consumes just around 20 mA of current. However, the radio operates on a low-duty cycle, and spends most of its time in 'sleep mode.' In cases where Tethercell is used merely for switching on or off devices, it will be able to run for approximately six to twelve months on a single AAA-size battery. However, when it is configured to do more tasks and the host device draws more power, the battery might need replacement earlier. Tethercell utilises Bluetooth 4.0 – the new low-power Bluetooth protocol that is expected to play a huge role in the Internet of Things. It has a range of around 18 metres (60 feet) indoors and 30 metres (100 feet) in open grounds.

Users will be able to tailor the behaviour of their Tethercells using the Tetherboard app with simple iOS-style app controls. An app developer kit can also be downloaded for free to build your own applications. The device also takes care of your privacy by including password protection.



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